

APPENDIX 7.1 ADT SORTED LIST SPECIFICATION

ADT Sorted List

A sorted list is a linear collection of entries in sorted order.

add(T newEntry)

Description : Adds **newEntry** to the sorted list such that the list remains sorted.
Postcondition : **newEntry** has been added to the sorted list in a position which ensures that the list remains sorted .

boolean remove(T anEntry)

Description : Removes the first or only occurrence of **anEntry** from the sorted list.
Postcondition : **anEntry** has been removed from the list. If **anEntry** was not located in the list, the list remains unchanged.
Returns : true if **anEntry** was located and removed, else false.

clear()

Description : Removes all entries from the list.
Postcondition : The list is now empty.

boolean contains(T anEntry)

Description : Determines whether the list contains **anEntry**.
Postcondition : The list remains unchanged.
Returns : **true** if **anEntry** is in the list, or false if not.

int getNumberOfEntries()

Description : Gets the number of entries currently in the list.
Postcondition : The list remains unchanged.
Returns : The number of entries currently in the list.

boolean isEmpty()

Description : Determines whether the list is empty.
Postcondition : The list remains unchanged.
Returns : **true** if the list is empty, or false if not.